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CS5053NI– Cloud Computing & IoT

IOT-Based Flame and Gas Safety Device with Smart Emergency Response

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Group Members

London Met ID	Student Name
24*****56	Pankaj Rai
24*****54	Palden Lama
24*****49	Eijkeyal Pakhrin
24*****29	Suhana Shrestha
24*****58	Priyanka Kumari Shah
24*****79	Lucky Lawaju

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Abstract

Fire and gas leakage are serious safety concerns in residential, commercial, industrial environments and in almost all the places where human works, lives or visits. Many accidents occur due to delayed detection or lack of proper alert systems. Our project focuses on designing and developing an IOT-based Flame and Gas detection system using Arduino UNO to provide early warning and improve safety.

The system uses two sensors, a flame sensor and an MQ-series gas sensor. When flame is detected a warning message is displayed on LCD, buzzer is activated with a specific sound pattern and a motor will start pouring the water to stop the fire which can further accidents. When a gas is detected, a different message is shown on the LCD and the buzzer produces a different sound pattern. In the case where both the buzzer and gas are detected simultaneously, a critical warning message is displayed and the buzzer generates a continuous alarm sound and the motor will turn on as well.

Using different buzzer sounds allows users to quickly identify the type of danger even without looking at the display and using the motor helps to prevent the rapid increase of the fire until the help arrives. The system is low-cost, simple to implement and suitable for homes and small workplaces.

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1. Introduction

Internet Of things (IOT) is an important technology which allows devices to connect, collect data and perform automatic actions without human involvement. It is used widely in almost all sectors including homes, industries, agriculture, health, education and for security purposes. Safety is one of the most important among them. In context of Nepal, fire accidents and gas leakage incidents are most frequent, especially in kitchens where LPG cylinders are used. The realization of the danger only happens when it becomes too late. This shows the need for an early warning system that can detect fire or gas and respond quickly to prevent the damages and loss. so, we decided to develop a system that can detect both flame and gas leakage in this project.

This project focuses on developing an IOT-based flame and gas detection system that can detect dangerous situations early and provide alerts through visual, audio and mechanical responses.

1.1. Current scenario

It is rare that any Nepali households, restaurants and small industries have installed any kind of safety equipment like flame detectors or gas alarms. They usually depend on smell or visual signs but gas leakage or small flames are not always noticeable in time. Some of the common problems that we have seen are:

- Gas cylinder leak without anyone realizing
- Kitchen fires start due to unattended stoves
- Electrical sparks causing small flames
- Firefighting systems are expensive and not that practical for normal housings.

Such issues lead to accidents and cause many losses. Our system can help reduce these risks by detecting danger as early as possible taking action automatically.

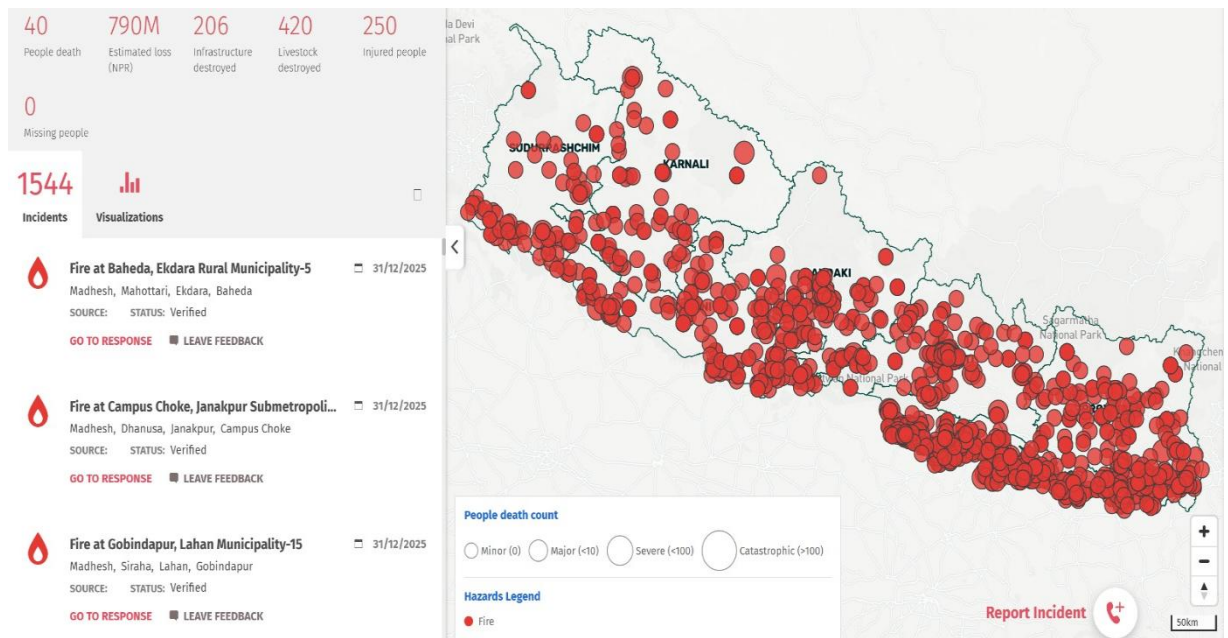


Figure 1 Fire incidents heat map of Nepal

Annual Occurrence

Figure 1: Number of fire, landslide and flood occurrence from 2016 to April,2025

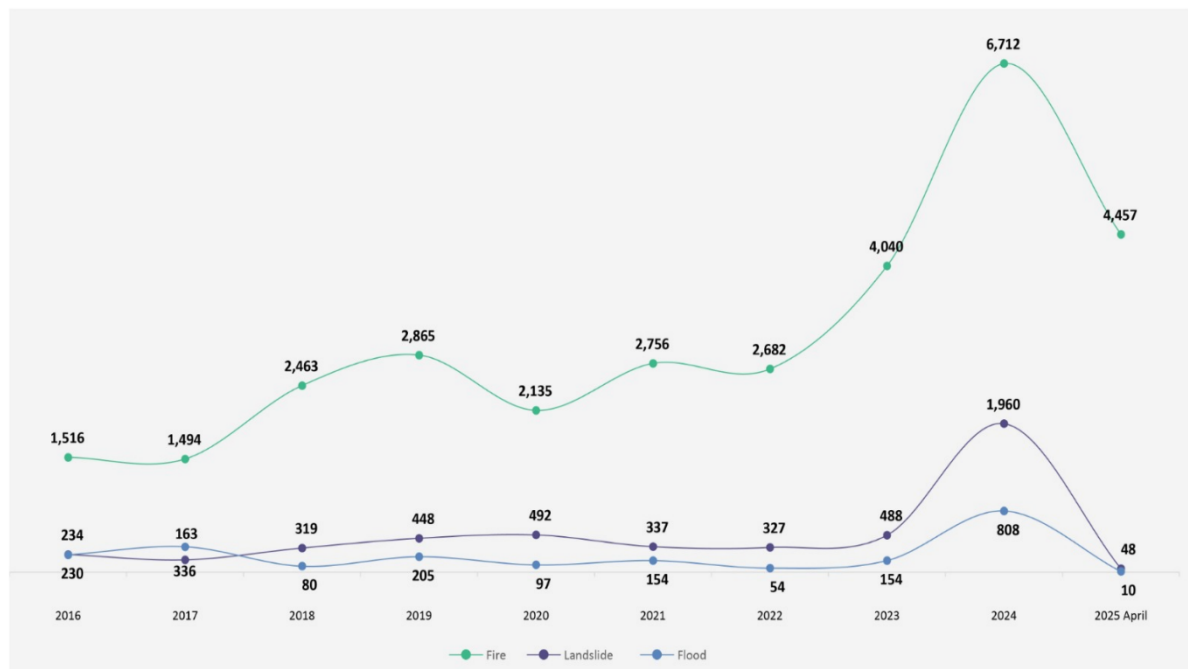


Figure 2 Compared Graph of Fire incidents from 2016 to 2025

1.2. Problem statement and project as a solution

Fire and gas leakage are common safety problems in homes and small workplaces. Gas leakage is difficult to notice early and fire can spread very fast if not detected on time. Most people still depend on smell or sight to identify these dangers, which is not always reliable. Although safety systems are available, many of them are expensive and do not clearly indicate the type of danger.

To solve this problem, our project depends on an IOT-based Flame and Gas detection system using Arduino. The system uses a flame sensor and a gas sensor to continuously monitor the environment. When flame is detected, a warning message is shown on the LCD, the buzzer sounds in a specific pattern and the motor turns ON. When gas is detected, the LCD shows Gas warning message and the buzzer sounds differently. If both flame and gas are detected, a critical warning message is displayed, the buzzer gives a continuous alarm and the motor is turned ON.

This system is simple, low-cost and helps users quickly understand the type of danger, improving safety and reducing risks.

1.3. Aims

Our project mainly aims to build an IOT-based flame and gas detection system that responds automatically and alerts users during dangerous situations.

1.4. Objectives

All the objectives to meet the aim of our project are listed below:

- ✓ To detect flame using flame sensor
- ✓ To automatically turn on water pump using relay
- ✓ To detect LPG/methane gas using a gas sensor
- ✓ To display warning messages on an LCD
- ✓ To activate buzzer in case of gas and flame detection
- ✓ To apply IOT concepts practically using Arduino and sensors
- ✓ To design a cost-effective and easy to use safety device

2. Background

The use of IOT in safety and automation is increasing widely because they reduce human efforts and give fast responses. Fire and gas leakage are two major hazards in Nepal, especially in residential areas and food businesses. Many people still rely on manual checking which is not always reliable. Gas leakage is difficult to detect manually and fire spreads rapidly if not controlled at an early stage. Many existing systems are either expensive or too complex for small users.

This creates the need for simple, affordable and effective detection system that can alert users and initiate basic safety actions automatically.

2.1. System Overview

The IOT-based flame and gas safety device is designed to provide early detection and quick response to fire and gas leakage incidents in homes and small workplaces. The main purpose of the system is to continuously monitor the environment and alert users immediately when a dangerous situation occurs.

The system uses Arduino UNO as the central controller, which receives input from flame sensor and an MQ-series gas sensor. These sensors continuously check for the presence of fire and gas in the surroundings area. When any abnormal condition is detected, the Arduino processes the data and activates the appropriate output devices.

To inform users clearly, the system uses an LCD display to show warning messages and a buzzer to produce different alert sounds depending on the type of hazard. In case of flame detection, the system also activates a water pump through a relay module to help reduce the spread of fire until further action is taken.

Overall, the system is simple, low-cost and easy to use. It demonstrates how IOT technology can be applied in real-life safety systems to reduce risks, improve response time and protect lives and property.

2.2. Hardware Architecture

Below is the sample setup of our hardware architecture with all the sensors and connections made as a system.

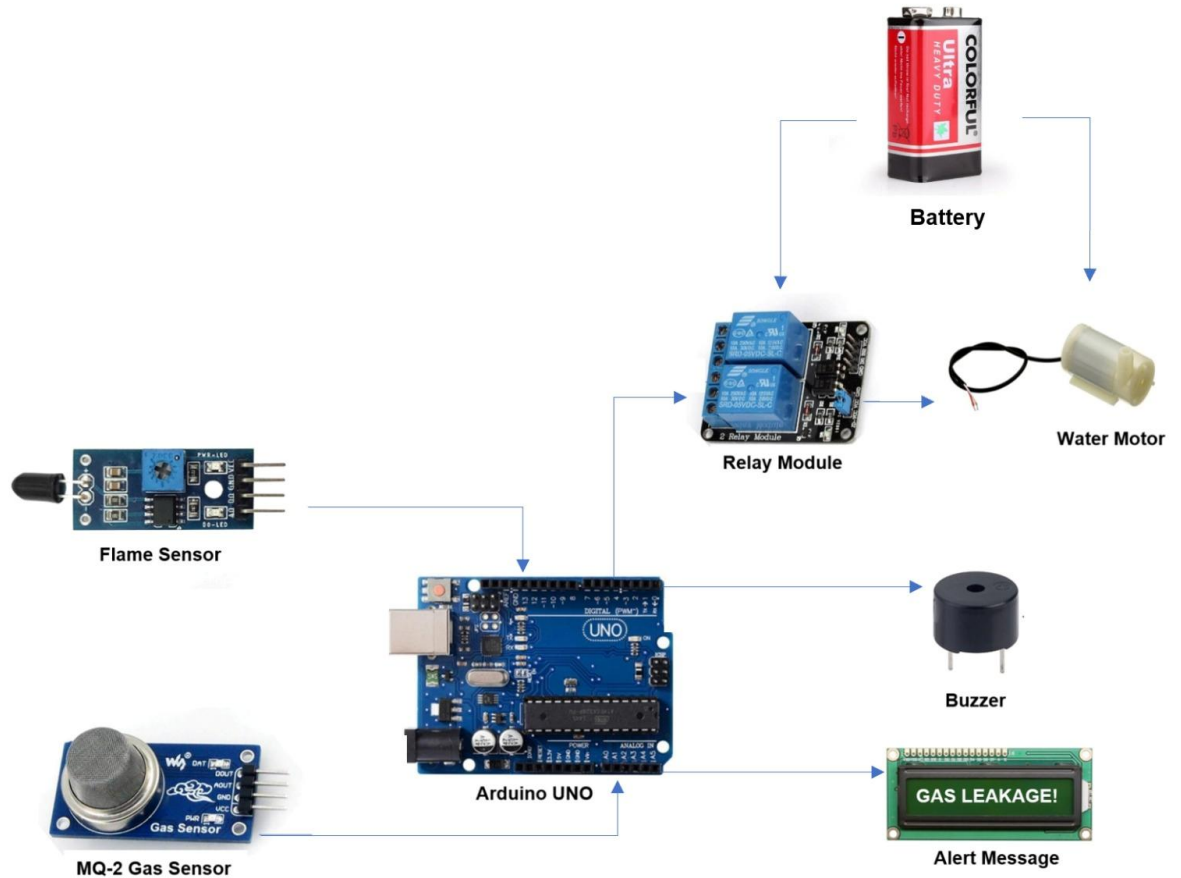


Figure 3 hardware architecture

2.3. Circuit Diagram

The circuit diagram of our system is drawn with the help of Tinker Cad and its image of the setup is presented below.

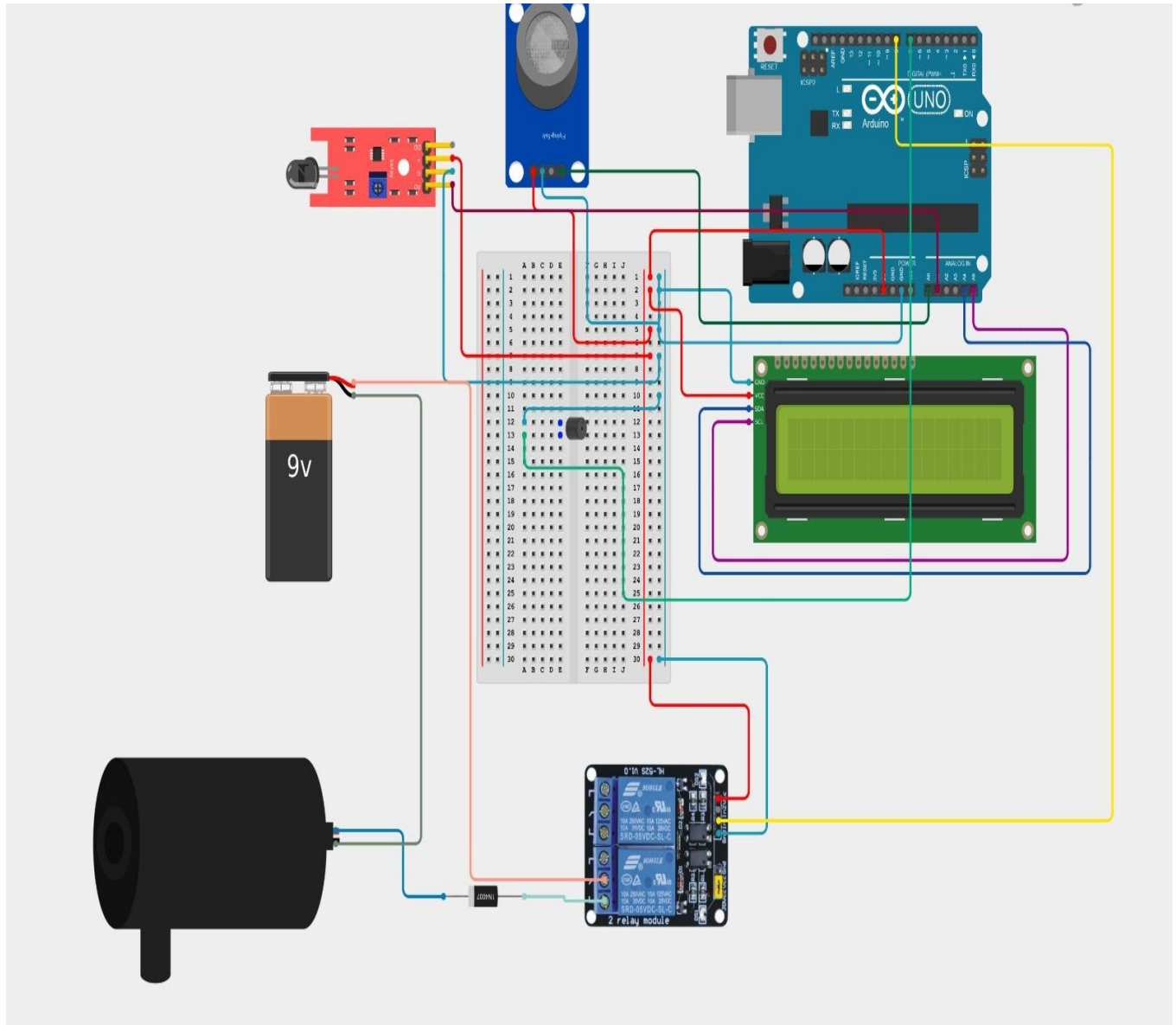


Figure 4 circuit diagram

2.4. Flowchart

The flowchart represents the step-by-step working mechanism and the functionality of our project which is easy to understand and interpret.

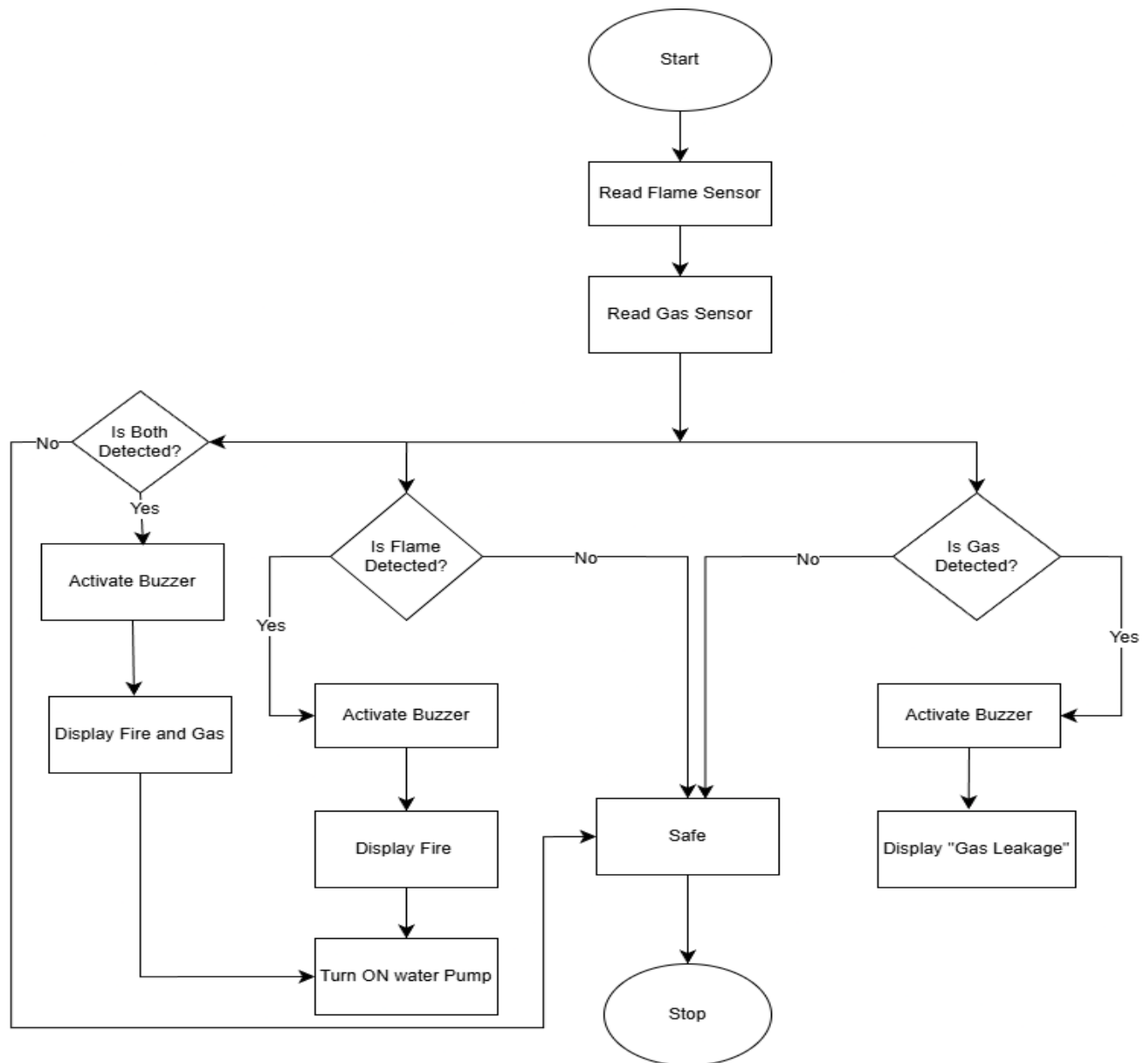


Figure 5 flowchart

2.5. Requirement Analysis

All the hardware and software that are required for our project are well researched and analysed. We picked the best devices and software that are efficient, reliable, simple and budget-friendly so that anyone can afford it for the safety.

2.5.1. Flame sensor

A sensor which is used to detect the properties of flames and trigger relevant actions is known as flame sensor. It is responsive to infrared (IR) radiation released by flames, which is converted into an electrical signal within a specific range (Winsen Sensors, 2025). In this project, flame sensor is designed to continuously receive environmental stimuli and send electrical signal to the microcontroller upon flame detection, which, simultaneously triggers the relay module to activate water pump. This allows for early identification and mitigation of blazing fire.

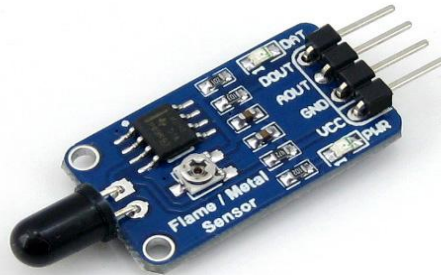


Figure 6: Image of Flame Sensor

2.5.2. MQ Gas Sensor

Methane/CH₄ Quality (MQ) gas sensor is used to detect the characteristics of combustible gases such as LPG and methane. It is designed to produce analog output corresponding to gas concentration released in air (Soldered Electronics, 2025). In this project, gas sensor is devised to continuously monitor the gas

intensity in air and send analog signal to the microcontroller. Upon the exceeding level of gas leakage, the system warns the user by activating a buzzer. This allows for awareness and immediate precautions of gaseous hazards.



Figure 7: Image of Gas Sensor

2.5.3. Arduino UNO

An open-source prototyping platform which is based upon Atmega328 microcontroller to build electronic projects with well-defined hardware, IDE and user-friendly programming languages is known as Arduino UNO (Robocraze, 2022). In this project, this tool performs as the central unit for controlling flame and gas sensor-based system. It continuously reads sensor data; when flame sensor detects fire, Arduino UNO activates relay module and manages water pump functionality. Simultaneously, when the gas sensor detects excessive level of gas density, Arduino UNO triggers the buzzer.



Figure 8: Logo of Arduino UNO

2.5.4. Relay Module

An interface used as a switching device to regulate low-power supply with a high-power supply circuit without incorporating it on same electrical application to safeguard delicate components is known as relay module (Schneider Electric, 2025). In project, relay module is used as an electric switch between low-power supply circuit, Arduino UNO and high-power device, water pump. Upon flame detection, the Arduino transmits an electrical signal to the relay module, which then activates the water pump for fire mitigation.

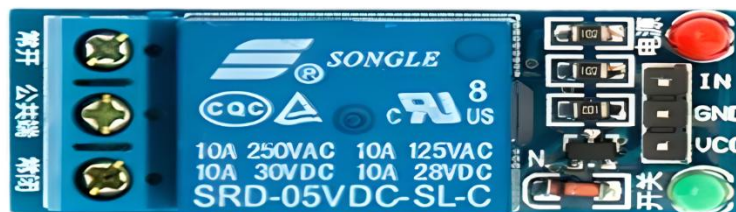


Figure 9: Image of Relay Module.

2.5.5. Water Pump

A device which controls the water flow to eradicate the risk of fire upon flame detection is known as water pump. In this IoT project, this mechanical tool remains crucial in mitigating fire hazards in home and industrial sector by increasing safety. When a flame is detected, an electrical signal transmitted to the relay module automatically activates the water pump and supplies water to impacted areas, minimizing it from spreading.



Figure 10 water pump

2.5.6. Buzzer

A component which gives digital output by producing a sound to alert users is known as buzzer. In this IoT project, the functionality of a buzzer notifies user when a flame or gas is detected. The Arduino sends signal to the buzzer when flame within a specific range and gas concentration above a suitable level is detected. This generates an alarming sound which makes the surrounding aware of possible hazard.



Figure 11: Image of buzzer

2.5.7. Breadboard

A prototyping tool which is used to hold and connect electronic components without soldering is known as a breadboard. It establishes connection through conductive metal strips situated below the plastic board (Robocraze, 2022). In this project, breadboard acts as a core component to build circuit connections among sensors, Arduino UNO and utilities, allowing flexible electrical pathways.

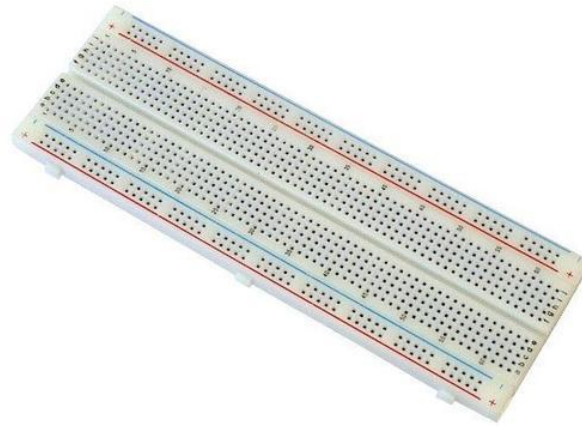


Figure 12: Image of Breadboard.

3. Development

The development part of our project includes the overall development process of the IOT-based Flame and Gas safety device. It describes how the idea was formed, how the system was planned and designed, the resources collected and how the system was implemented, tested, debugged and finalized.

3.1. Brainstorming and group discussion

At the initial phase of our project, we conducted the group discussion and brainstorming sessions to identify real-life problems related to safety that could be solved using IOT concepts. During the discussion, fire accidents and gas leakage incidents were identified as common and serious problems, especially in households, restaurants and small workplaces in Nepal. Each group member shared ideas and experiences related to gas leakage and fire hazards. And after evaluation, our group decided to develop a flame and gas detection system rather than focusing on a single hazard.

3.2. Planning and designing

After finalizing the idea, the system was planned and designed. We selected Arduino UNO as the main controller, while flame and gas sensor were chosen as input devices. Output components such as buzzer, LCD, relay module and water pump were added for alert and response. Apart from that, design diagrams were prepared to clearly understand the system structure and working process.

3.3. Resource collection

Once the design was complete, the required hardware and software components were collected. Most of the components were low-cost and easily available. We worked together to arrange sensors, Arduino, relay module, LCD, buzzer, water pump and jumper wires. The Arduino IDE was used for coding and uploading the program.

3.4. System development

We did the system development step by step. At first, the Arduino UNO was connected to the computer for power supply and programming. Then the breadboard was used to connect sensors and output devices. The flame sensor and gas sensor were connected to detect the fire and gas leakage. The LCD and buzzer were used to provide warnings, while the relay module controlled the water pump during fire detection.

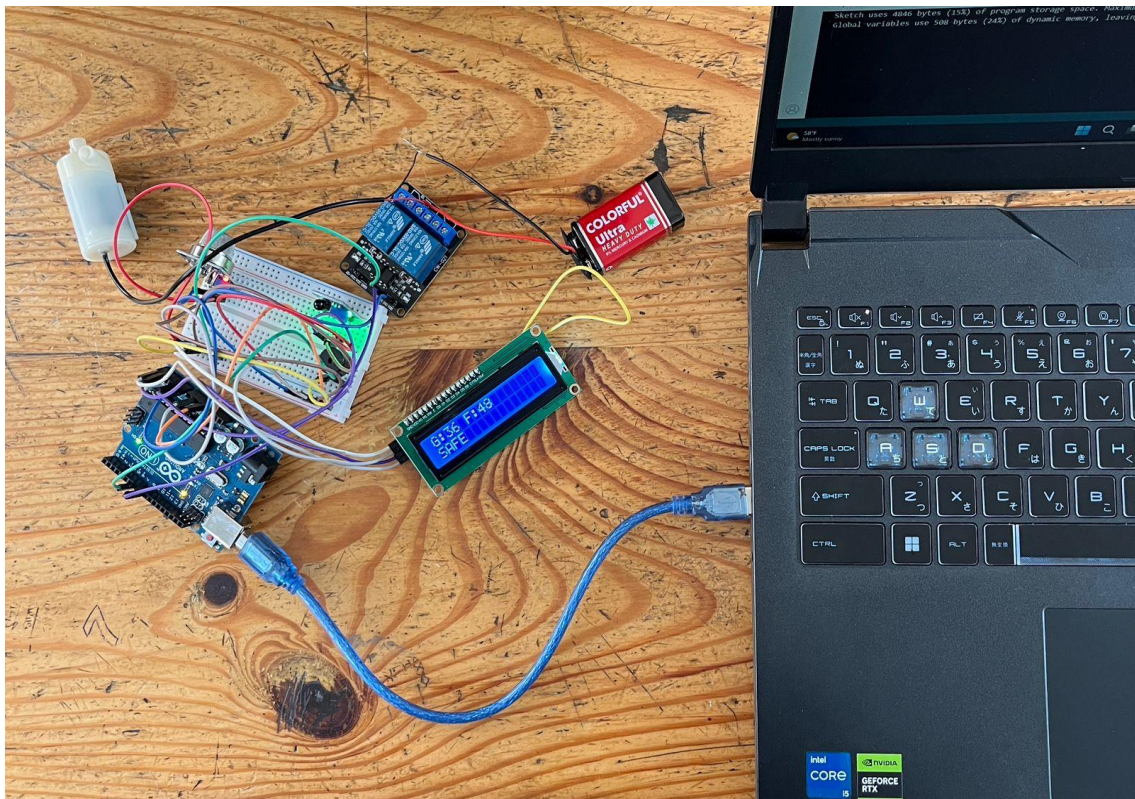


Figure 13 Connection setups of sensors, breadboards and laptop

3.5. Implementation

After completing hardware connections, we implemented the system using programming. The system continuously reads sensor values and responds accordingly. Different buzzer sounds and LCD messages were programmed to indicate flame detection, gas detection and critical situations. The water pump automatically turns on when fire is detected.

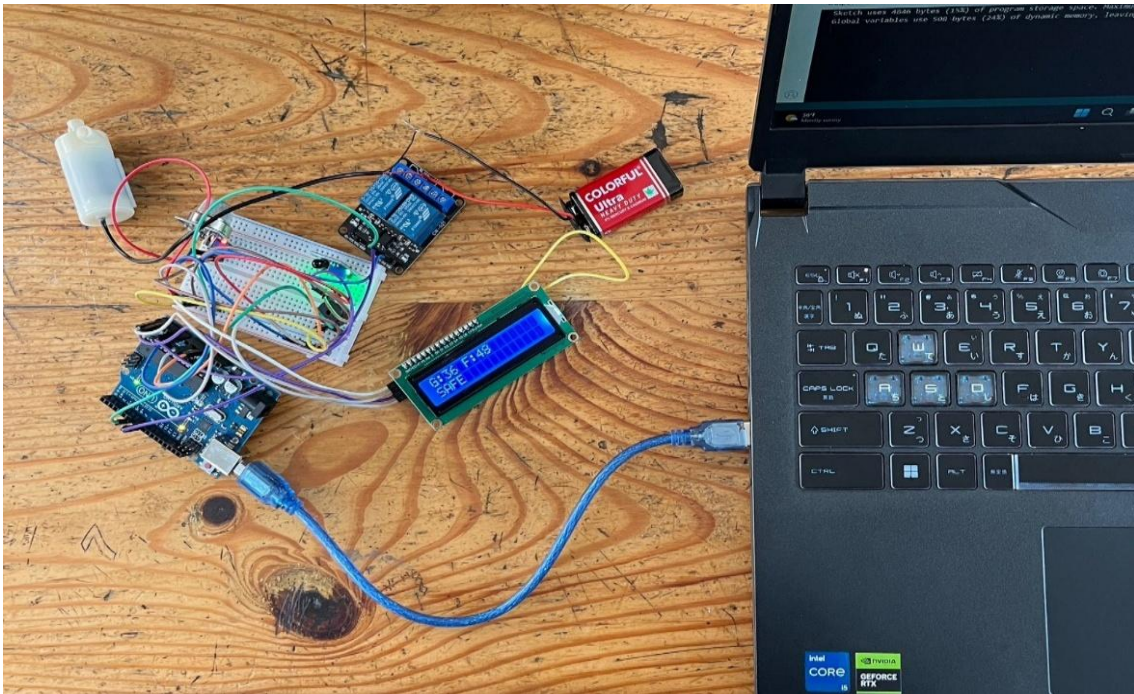


Figure 14 gas detection message

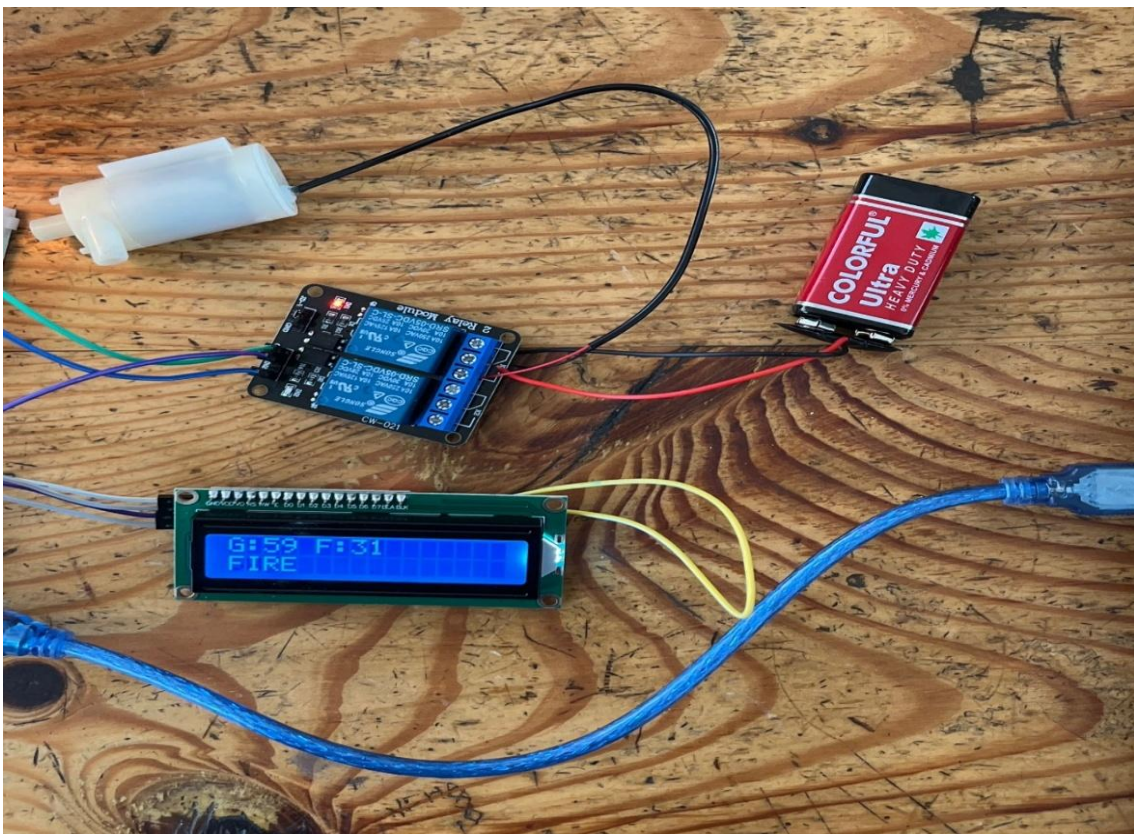


Figure 15 fire detection message

3.6. Testing

After all the setups and connections, we tested the system under different conditions to check accuracy and reliability. We tested flame detection, gas detection and combined detection scenarios. The system produced correct buzzer alerts, displayed appropriate messages and activated the water pump when required.

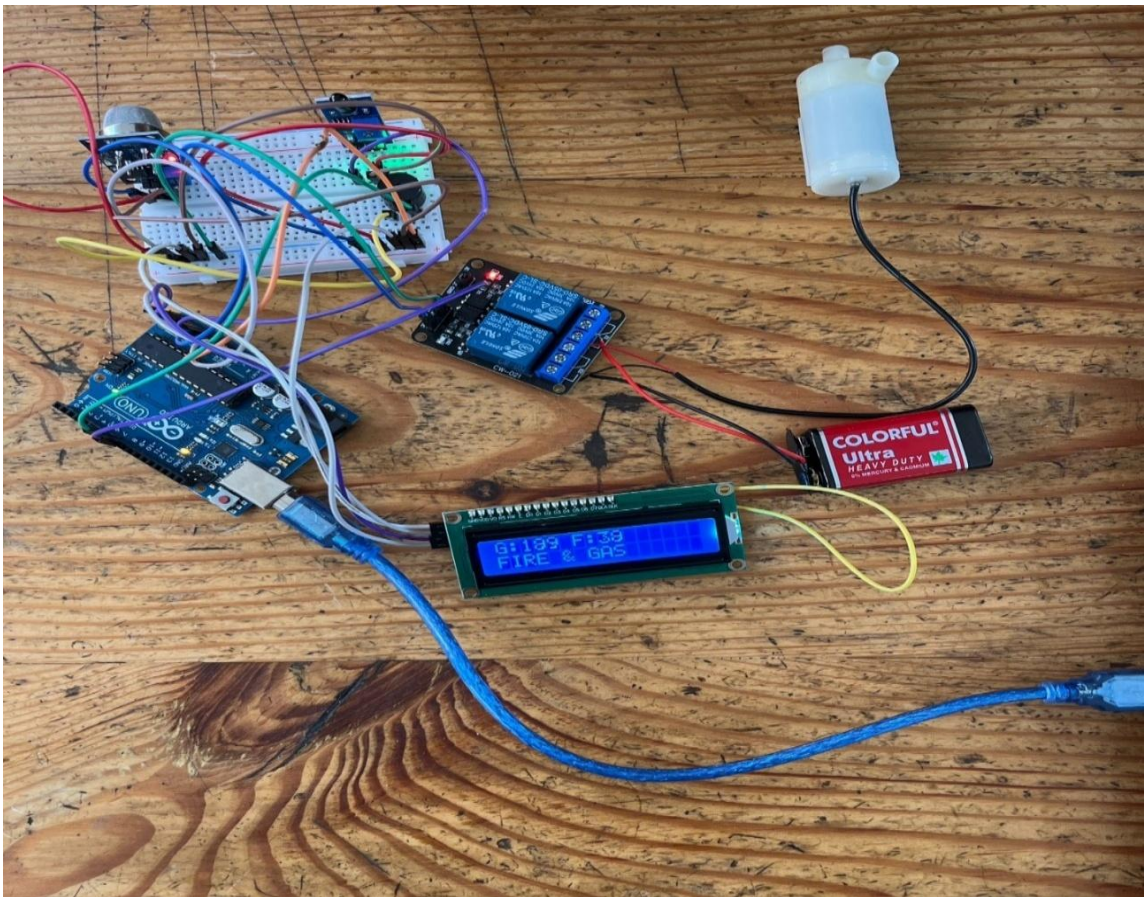


Figure 16 detecting both fire and gas

3.7. Debugging and finalizing

We faced some minor issues such as sensor sensitivity and wiring problems during the testing. We fixed them by adjusting values in codes and rechecking the connections. After debugging, the system worked and our group approved the system and finalized it successfully.

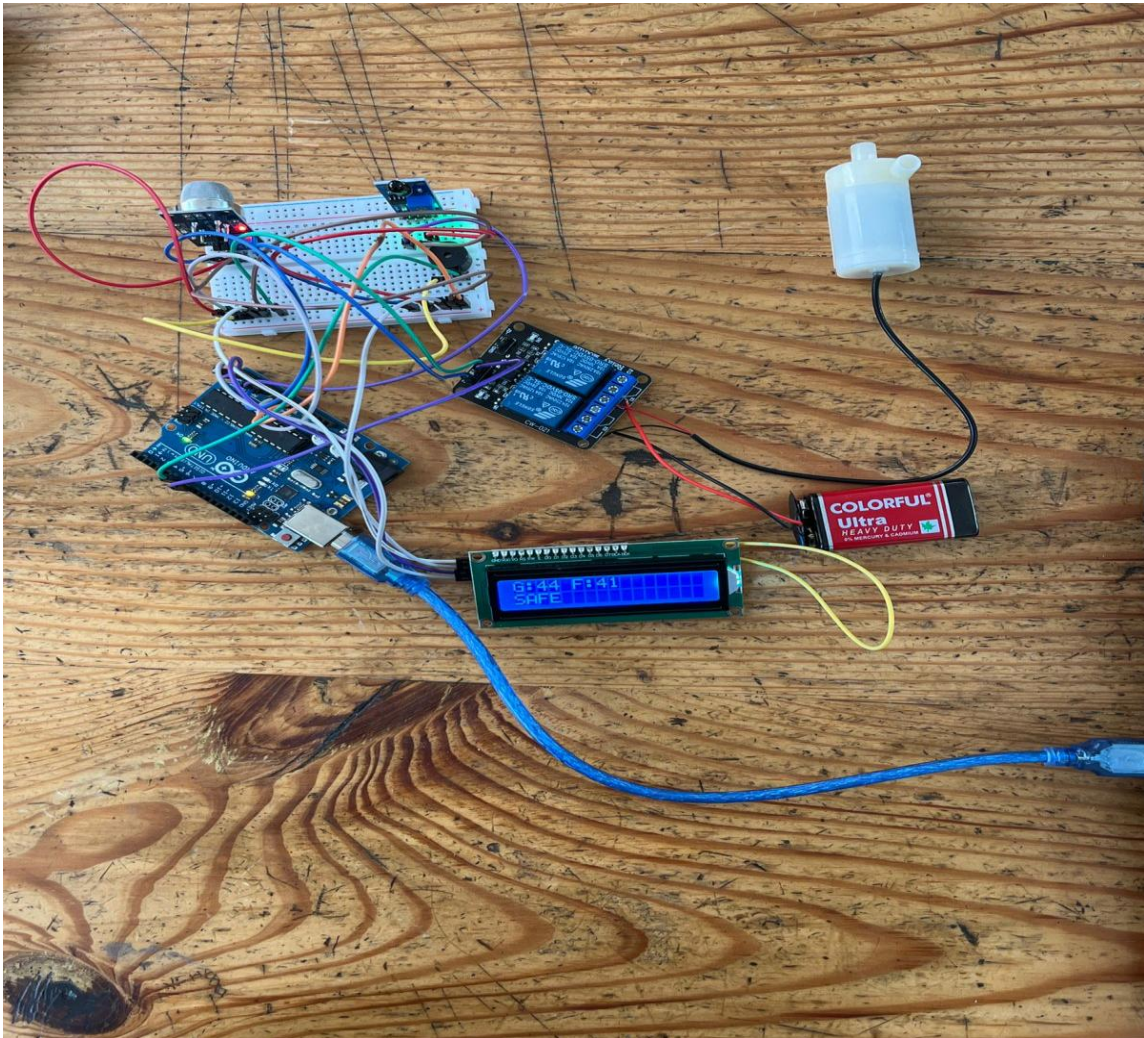


Figure 17 final working prototype

4. Results and Findings

4.1. Results

The final system successfully demonstrates the working of an IOT-based flame and gas safety device with smart emergency response. The system is now able to continuously monitor the surround environment and accurately detect flame and gas leakage.

When flame was detected, the system displayed a warning message on the LCD, activated the buzzer with a specific alarm pattern and automatically turned on the water pump using the relay module. When gas was detected, a different warning message and buzzer sound were produced. In critical conditions where both flame and gas were detected simultaneously, the system generated a continuous alarm and activated the water pump.

Overall, the system worked reliably and met all the objectives defined at the beginning of the project.

4.2. Findings

We carried out several test cases to verify the performance of the system and the findings from each test are described below.

4.2.1. Test 1

Table 1 Flame detection testing

Test NO:	1) flame detection
Objective:	To test the response of the system when flame is detected.
Action:	A flame source was brought near the flame sensor.
Expected Result:	LCD should display flame warning, buzzer should activate and water pump should turn ON.
Actual Result:	The system detected the flame correctly, displayed the warning message, activated the buzzer and turned on the water pump.
Conclusion:	Test was successful.

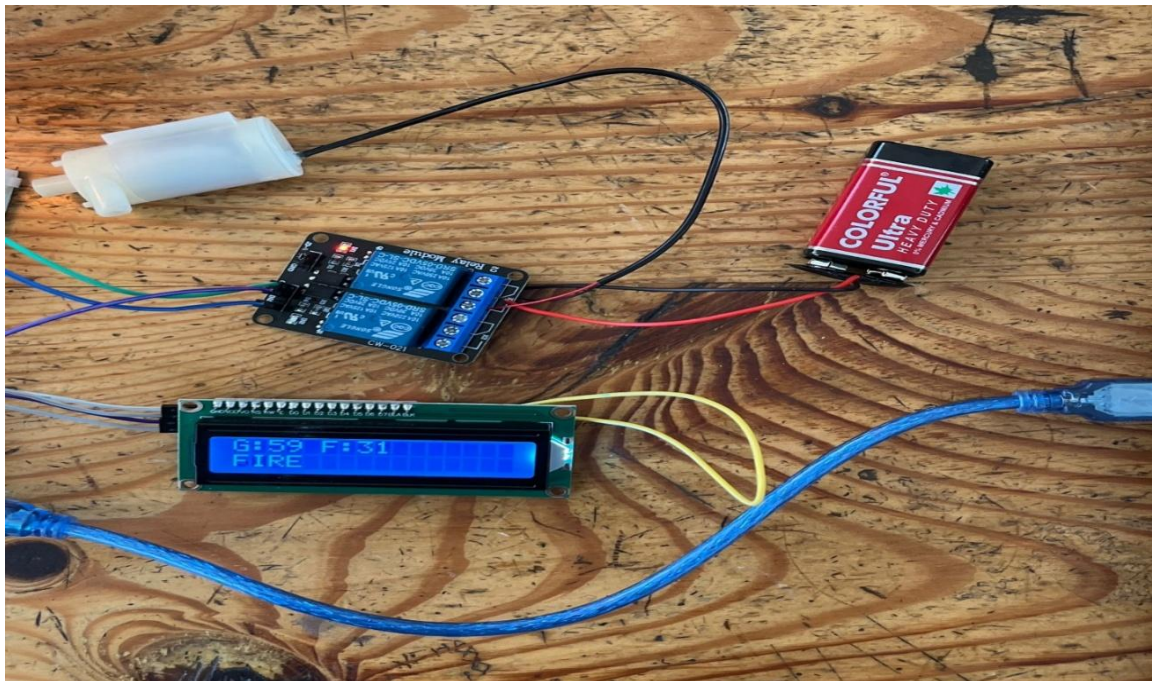


Figure 18 flame detection testing

4.2.2. Test 2

Table 2 Gas detection testing

Test NO:	2) Gas detection
Objective:	To test gas leakage detection.
Action:	Gas was released near the gas sensor.
Expected Result:	LCD should display gas warning and buzzer should produce gas alert sound.
Actual Result:	Gas was detected successfully, correct warning message was displayed and buzzer alert was triggered.
Conclusion:	Test was successful.

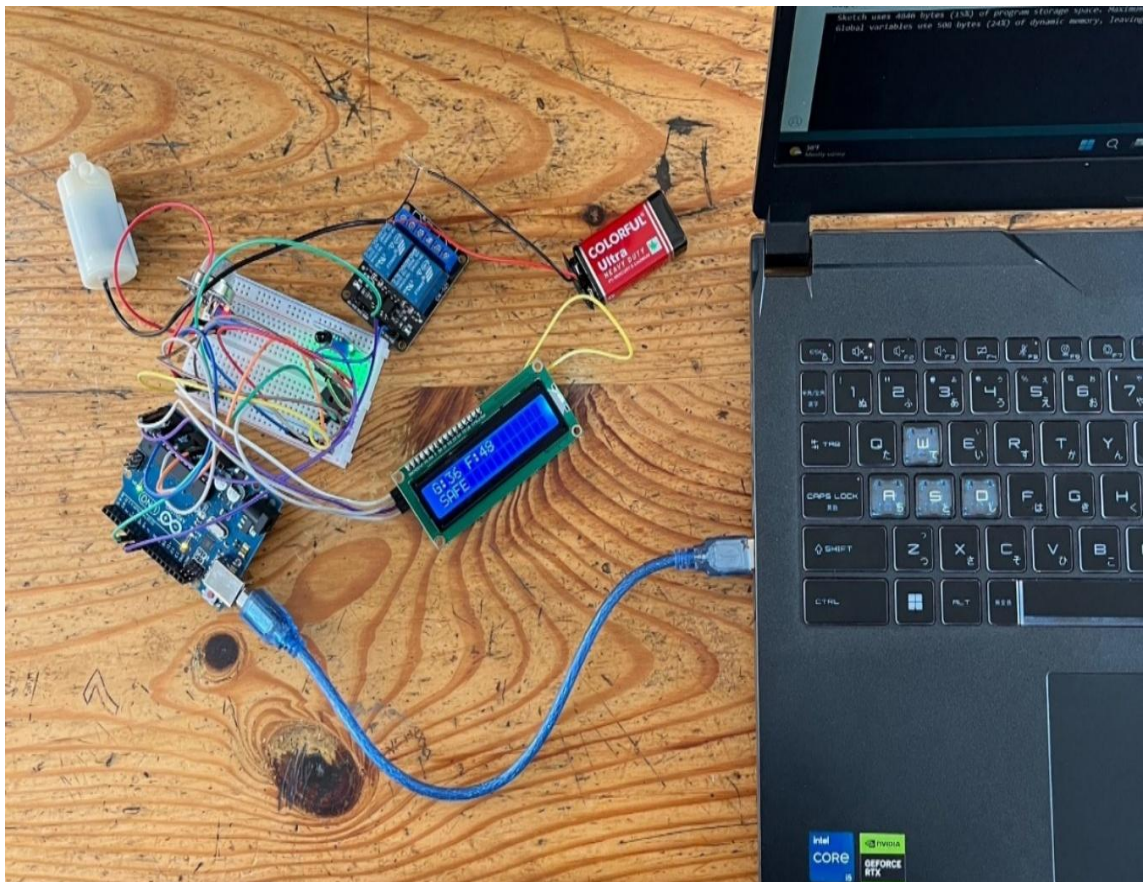


Figure 19 gas detection testing

4.2.3. Test 3

Table 3 Combined flame and gas detection testing

Test NO:	3) Combined Flame and Gas detection
Objective:	To test system response during a critical situation.
Action:	Both flame and gas were introduced near the sensors.
Expected Result:	System should display critical warning, buzzer should sound continuously and water pump should turn ON.
Actual Result:	The system correctly identified the critical condition and performed all emergency action.
Conclusion:	Test was successful.

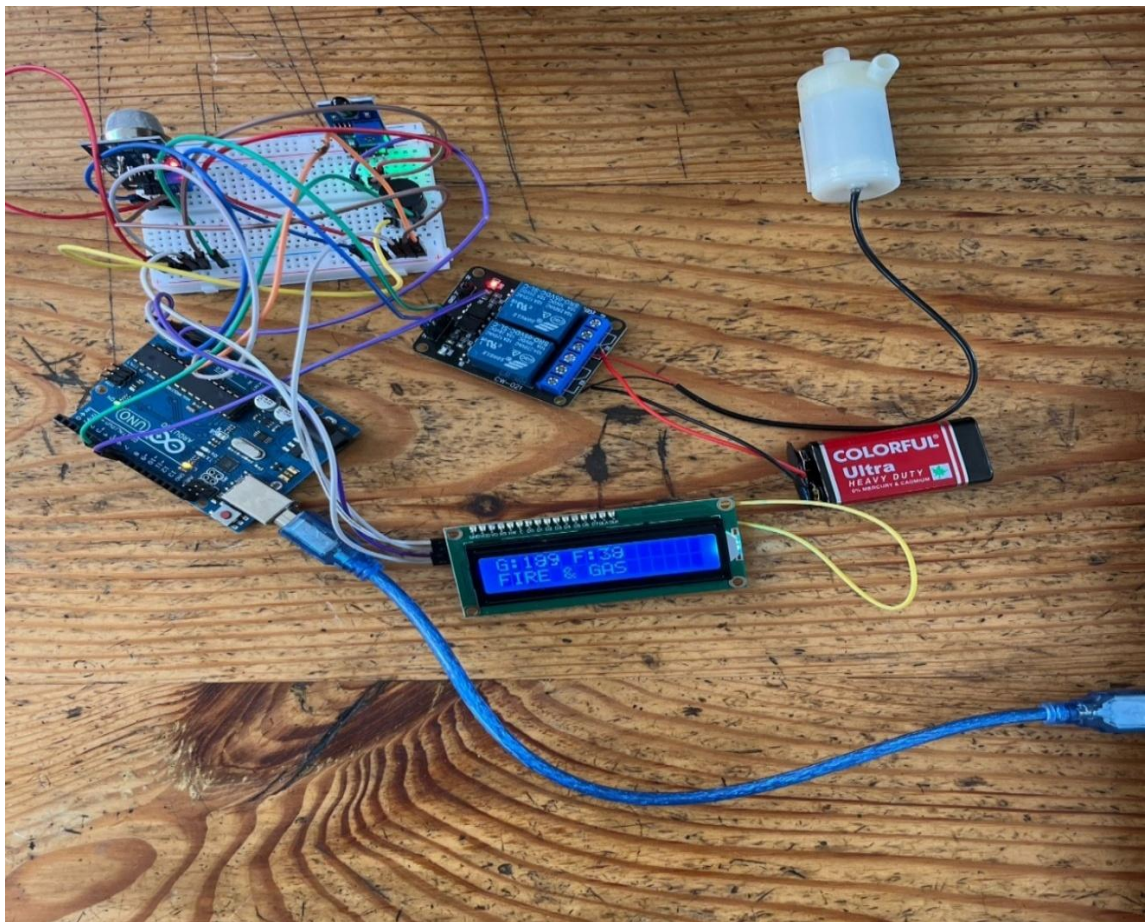


Figure 20 combined gas and flame detection testing

4.2.4. Test 4

Table 4 Buzzer and functionality testing

Test NO:	4) Buzzer and LCD functionality
Objective:	To verify buzzer sound and LCD message accuracy.
Action:	Different hazard scenarios were created.
Expected Result:	Different messages and buzzer patterns for flame, gas and critical conditions.
Actual Result:	The system displayed correct messages and produced distinct buzzer sounds.
Conclusion:	Test was successful.

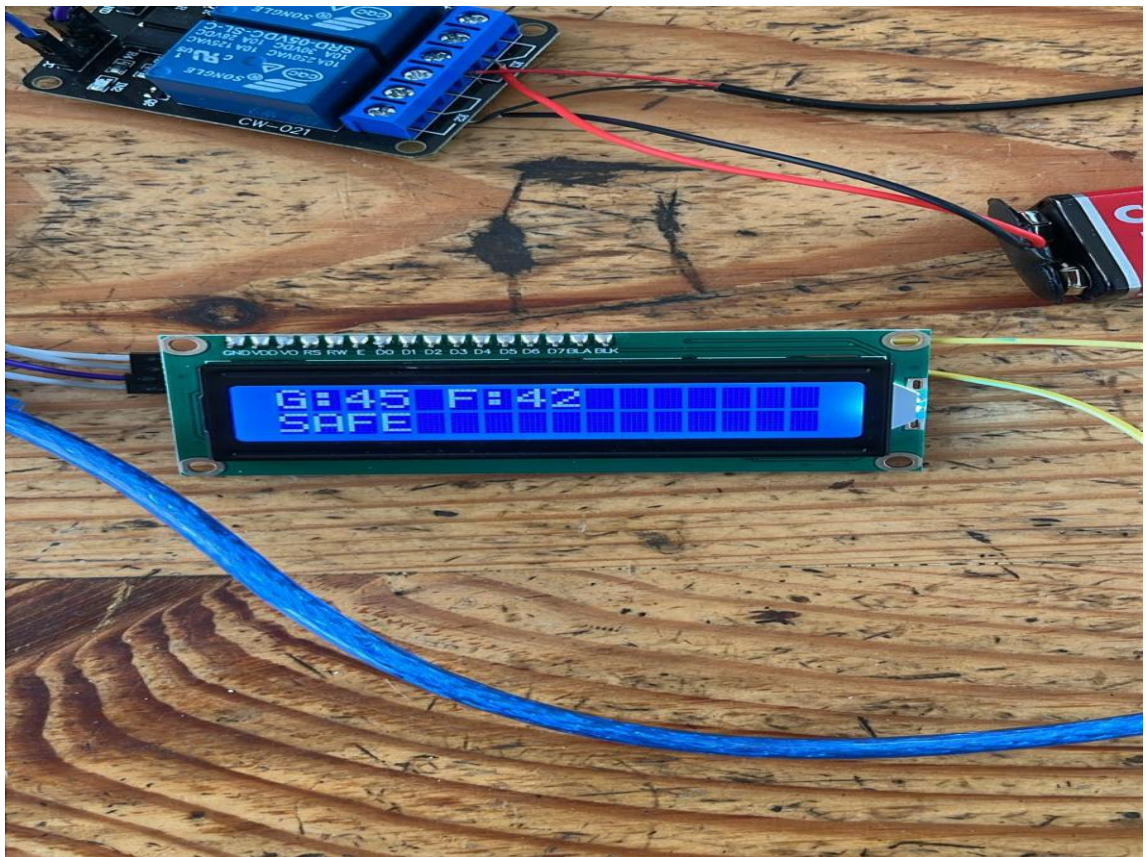


Figure 21 LCD functionality testing

4.2.5. Test 5

Table 5 Relay and water pump operation testing

Test NO:	5) Relay and Water Pump operation
Objective:	To test automatic water pump activation.
Action:	Flame detection was triggered.
Expected Result:	Relay should activate the water pump.
Actual Result:	Relay operated correctly and water pump was activated.
Conclusion:	Test was successful.

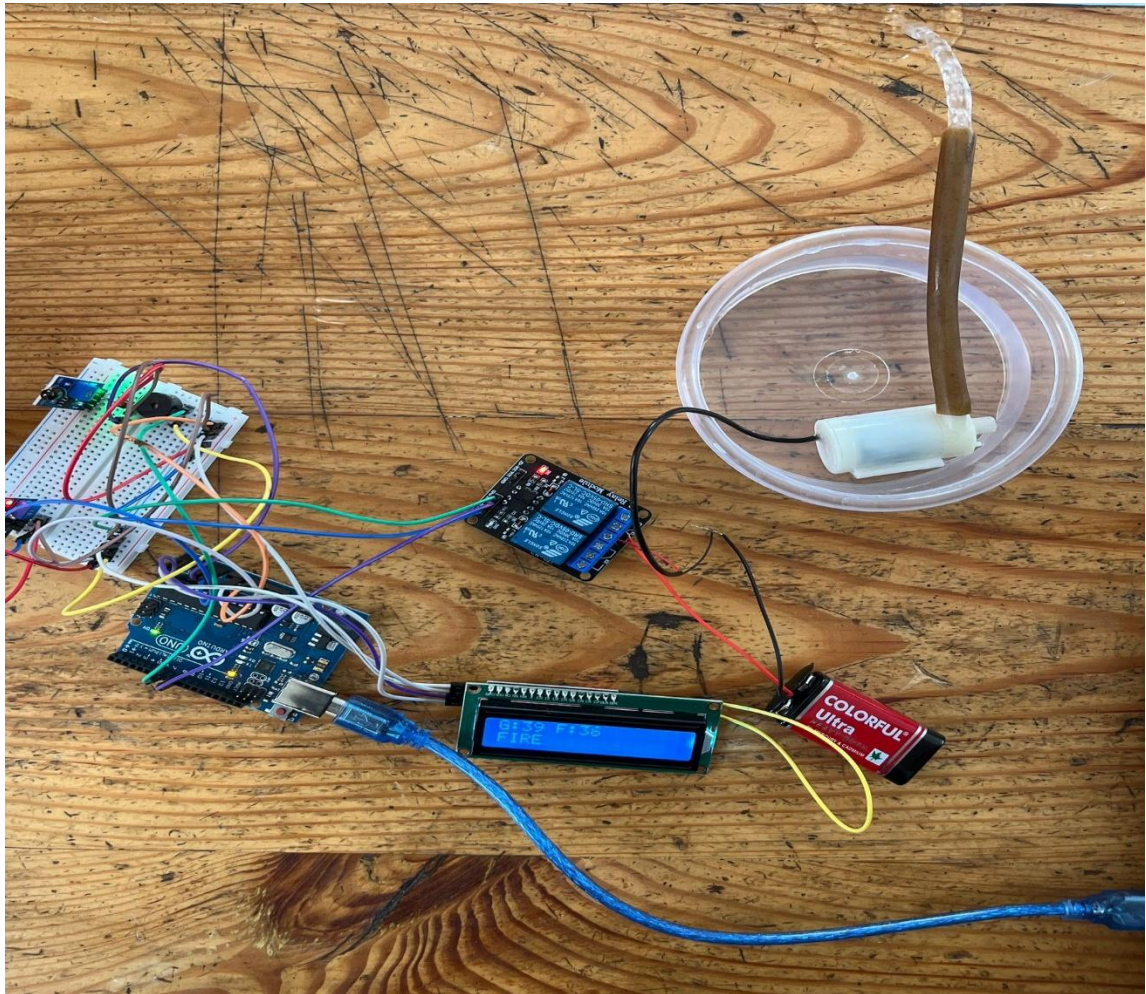


Figure 22 water motor testing

5. Future Works

Our current IOT-based Flame and Gas Safety Device work effectively for detecting flame and gas leakage and providing immediate alerts. However, there are several improvements that can be made in the future to enhance the system functionalities.

In future versions, the system can be integrated with Wi-Fi or GSM modules so that alert messages can be sent directly to user's mobile phones. This would be especially useful when users are not physically present near the system. Mobile application or cloud-based monitoring can also be added to view real-time sensor data remotely.

Additional safety features such as automatic gas valve control can be included to stop gas supply during leakage. The system can also be expanded by adding smoke and temperature sensors for more accurate fire detection. With better casing and power backup, the device can be made suitable for real-world installation in homes, restaurants and small industries.

6. Conclusion

Our project successfully demonstrated the practical use of Internet of Things (IOT) concepts by developing an IOT-Based Flame and Gas Safety Device with smart emergency response. The system was designed to detect flame and gas leakage at an early stage and respond automatically using visual, audio and mechanical alerts.

This project helped us gain hands-on experience and practical knowledge in working with Arduino, sensors and real-time system integration. It also improved skills such as teamwork, problem-solving, time management, idea sharing and communication. The final system that we built met all the project objectives and proved to be cost-effective and reliable safety solution.

In a nutshell, our project shows how simple IOT-based systems can play an important role in improving safety and preventing accidents in everyday environments.

7. References

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8. Appendix

Code

```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

#define GAS_ANALOG    A0
#define FLAME_ANALOG  A1
#define BUZZER_PIN    7
#define RELAY_PIN     8

#define GAS_THRESHOLD  150
#define FLAME_THRESHOLD 40

LiquidCrystal_I2C lcd (0x27, 16, 2);

void setup () {
    pinMode(BUZZER_PIN, OUTPUT);
    pinMode(RELAY_PIN, OUTPUT);

    digitalWrite(RELAY_PIN, HIGH);

    lcd.init();
```

```
lcd.backlight();

Serial.begin(9600);


lcd.print("Initializing");

delay (1500);

lcd.clear();

}


void loop () {


int gasValue  = analogRead(GAS_ANALOG);

int flameValue = analogRead(FLAME_ANALOG);


bool gasDetected  = gasValue > GAS_THRESHOLD;

bool flameDetected = flameValue < FLAME_THRESHOLD; // low = fire


lcd.setCursor(0, 0);

lcd.print("G:");

lcd.print(gasValue);

lcd.print(" F:");

lcd.print(flameValue);

lcd.print("  ");
```



```
lcd.setCursor(0, 1);  
  
lcd.setCursor(0, 1);  
  
if (flameDetected && gasDetected) {  
    lcd.print("FIRE & GAS ");  
} else if (flameDetected) {  
    lcd.print("FIRE ");  
} else if (gasDetected) {  
    lcd.print("GAS ");  
} else {  
    lcd.print("SAFE ");  
}
```

```
if (flameDetected && gasDetected) {  
    digitalWrite(BUZZER_PIN, HIGH);  
}  
  
else if (flameDetected) {  
    digitalWrite(BUZZER_PIN, HIGH);  
    delay (100);  
    digitalWrite(BUZZER_PIN, LOW);  
    delay (100);  
}
```

```
else if (gasDetected) {  
    digitalWrite(BUZZER_PIN, HIGH);  
    delay (200);  
    digitalWrite(BUZZER_PIN, LOW);  
    delay (200);  
}  
else {  
    digitalWrite(BUZZER_PIN, LOW);  
    delay (500);  
}  
  
if (flameDetected && gasDetected) {  
    digitalWrite(RELAY_PIN, LOW);  
}  
else if(flameDetected){  
    digitalWrite(RELAY_PIN, LOW);  
}  
else {  
    digitalWrite(RELAY_PIN, HIGH);  
}  
}
```